Data Flow Diagram

Date	05 July 2026
Team ID	
Project Name	Green-Predict-Pro
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Farmer, Weather API).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Input, Predict Crop).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Crop Dataset, User Database).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Green-Predict-Pro DFD (Level 1)

External Entity: Farmer / Researcher → P1. Validate Input → P2. Feature Preprocessing → P3. ML Model Prediction (Random Forest Classifier) → P4. Recommendation Formatter → Farmer / Researcher.

Data Stores:

D1: Crop Recommendation Dataset (N, P, K, temperature, humidity, pH, rainfall, label).

D2: Trained Model Artifacts (model.pkl, scaler.pkl).

D3: Prediction History (user_id, inputs, predicted_crop, timestamp).

Data Flows:

- Farmer → P1: {N, P, K, temperature, humidity, pH, rainfall}
- P1 → P2: validated_inputs
- P2 → P3: scaled_feature_vector
- D1 → P3: training_data (during model build)
- D2 → P3: model & scaler artifacts
- P3 → P4: {predicted_crop, confidence}
- P4 → Farmer: recommended_crop + suitability_report
- P4 → D3: session log